

Proof of Funds: Implementation Brief

Planned KYT / AML graph scoring prototype for incoming Bitcoin transactions.



Informatics

PROJECT BRIEF

Item	Description
Context	Cryptoasset service providers must quantify risk for incoming transactions as part of AML / KYT procedures.
Questions	How can we quantify risk, for example by distance from darknet markets? How can scores be precomputed for large graphs?
Tasks	Define risk metrics, compute exposure per entity, and analyze the resulting risk distribution.

Objective

We will build a Bitcoin Proof of Funds prototype for KYT / AML screening. The system will start from known high-risk addresses, crawl their transaction neighbourhood, and assign each address an explainable risk score.

DATA WE FOUND

Data	Use in the prototype
GraphSense TagPacks	Open address labels for ransomware, darknet markets, scams, hacks, exchanges, miners, and services.
Seed addresses	6 tagged Bitcoin addresses: Locky ransomware, investment / gift-BTC scams, and the 2016 Bitfinex hack.
mempool.space API	Public transaction data for crawling the local neighbourhood around the seeds.

CORE APPROACH

Step	What we will do	Output
1. Label risk sources	Use GraphSense TagPacks to identify addresses linked to ransomware, darknet markets, hacks, scams, and services.	Tagged seed addresses
2. Build local graph	Fetch transactions around the seeds and convert value flow into a weighted directed graph.	Address graph
3. Score exposure	Combine graph distance, direct tainted value, and multi-hop taint propagation.	0-100 risk score
4. Analyze results	Export the score table and inspect the distribution in a notebook.	Case study + plots

RISK METRICS

Metric	Meaning
Distance	How many hops separate an address from known high-risk funds.
Direct exposure	How much incoming value comes directly from risky predecessors.
Haircut taint	How taint propagates through the graph over multiple hops.
Aggregate score	A weighted 0-100 score used to prioritize manual review.

Design principle

The score should be explainable. A compliance analyst should be able to see whether risk comes from proximity, direct value exposure, or indirect taint.

WHAT WE WILL DELIVER

Deliverable	Purpose
Python package	Reusable implementation for loading labels, building graphs, and computing risk metrics.
Notebook	Readable explanation of the pipeline and score distribution.
Risk table	One row per address with metric columns and final score.

SCOPE BOUNDARIES

Boundary	Reason
Bitcoin only	Keeps the first prototype focused on one transaction model and one data source.
Bounded crawl	Makes the demo reproducible, but may miss distant taint paths.
No full clustering	Change-address and entity clustering heuristics are future work.
Prototype weights	Severity weights are illustrative and not production policy.